King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 208 Major Exam I Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The general solution of the linear differential equation

$$\frac{dy}{dx} + 3y = 2xe^{-3x}$$

is given by

(a) $y(x) = e^{3x}\left(x^3 + c\right)$

(b) $y(x) = e^{-3x}\left(x^3 + c\right)$

(c) $y(x) = e^{-3x}\left(x^2 + c\right)$

(d) $y(x) = e^{3x}\left(2x^2 + c\right)$

(e) $y(x) = e^{3x}\left(x^2 + c\right)$

2. If $y = f(x)$ is the solution of the initial value problem

$$\frac{dy}{dx} = \frac{20}{x^2 + 16}, \qquad y(0) = 0,$$

then $y(4) =$

(a) $\dfrac{\pi}{4}$

(b) 0

(c) $\dfrac{7\pi}{4}$

(d) $\dfrac{5\pi}{4}$

(e) $\dfrac{3\pi}{4}$

3. A particle is moving along a straight line with position function $x(t)$. If the acceleration of the particle is $a(t) = 50\sin(5t)$, the initial position is $x(0) = 8$, and the initial velocity is $v(0) = -10$, then $x\left(\dfrac{\pi}{2}\right) =$

(a) 6

(b) 12

(c) 8

(d) 10

(e) 4

4. If $y(x)$ is the solution of the initial value problem

$$x\frac{dy}{dx} - y = 2x^2 y, \qquad y(1) = 1,$$

then $y(2) =$

(a) $2e^3$

(b) e^4

(c) 0

(d) $5e^3$

(e) $2e^4$

5. A certain city had a population of 25000 in 1960 and a population of 30,000 in 1970. Assume that its population will continue to grow exponentially at a constant rate. The population in 1980 is

(a) 38000

(b) 34000

(c) 48000

(d) 42000

(e) 36000

6. The matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$ is invertible if

(a) $c \neq 0,\ b \neq 0,\ c = b$

(b) $c \neq b,\ b \neq 0$

(c) $c \neq 0,\ b = 0,\ c \neq b$

(d) $c = 0,\ b \neq 0,\ c = b$

(e) $c \neq 0,\ b \neq 0,\ c \neq b$

7. Which one of the following differential equations is an exact equation?

(a) $(xy + e^y)\,dx + (xy + e^x)\,dy = 0$

(b) $(4x + y)\,dx + (6y - x)\,dy = 0$

(c) $(x^2 + 2xy^2)\,dx + (4xy + 2y^2)\,dy = 0$

(d) $\left(y^3 + \dfrac{x}{y}\right)dx + \left(3xy^2 - \dfrac{x^2}{2y^2}\right)dy = 0$

(e) $(\cos x + \ln y)\,dx + \left(\dfrac{x}{y} + \sin x\right)dy = 0$

8. If $y = x^m$ is a solution of the differential equation

$$x^2y'' + 5xy' + 4y = 0,$$

then $m =$

(a) 3

(b) 1

(c) -3

(d) -1

(e) -2

9. Given the system

$$\begin{aligned} 3x - y - 5z &= 3 \\ 4x - 4y - 3z &= -4 \\ x - 5z &= 2 \end{aligned}$$

Solve the system by using the Gaussian elimination method. (No other method is accepted.)

10. Verify that the differential equation

$$\left[\sin(x^2 + y) + 2x^2\cos(x^2 + y)\right] dx + \left[x\cos(x^2 + y) + 2ye^{y^2}\right] dy = 0$$

is exact and then solve it.

11. Solve the initial value problem

$$(x^2 + 1)\frac{dy}{dx} + 3x^3y = 6xe^{-\frac{3}{2}x^2}, \qquad y(0) = 1.$$

12. Let $A = \begin{bmatrix} -5 & -2 & 2 \\ 1 & 5 & -3 \\ 5 & -3 & 1 \end{bmatrix}$. Find the inverse of A only by using the adjoint formula.

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$\rho = e^{3x}$: $(e^{3x}y)' = 2x$, so $y = e^{-3x}(x^2 + c)$.
2	**(d)**	$y = 5\tan^{-1}(x/4)$, so $y(4) = 5 \cdot \frac{\pi}{4} = \frac{5\pi}{4}$.
3	**(a)**	$v = -10\cos 5t$, $x = -2\sin 5t + 8$, so $x(\pi/2) = 6$.
4	**(a)**	Separable: $y = Cxe^{x^2}$, $C = e^{-1}$, so $y(2) = 2e^3$.
5	**(e)**	$e^{10k} = \frac{6}{5}$, so $P(20) = 25000\left(\frac{6}{5}\right)^2 = 36000$.
6	**(e)**	$\det A = bc(c - b) \neq 0$, so $b \neq 0$, $c \neq 0$, $c \neq b$.
7	**(d)**	Only the first equation has $M_y = N_x = 3y^2 - \frac{x}{y^2}$.
8	**(e)**	Substitute $y = x^m$: $m^2 + 4m + 4 = (m + 2)^2 = 0$, so $m = -2$.
9	written	Row reduction gives $-23z = 0$, so $z = 0$, $y = 3$, $x = 2$: the solution is $(2, 3, 0)$.
10	written	Exact ($M_y = N_x$); $F = x\sin(x^2 + y) + e^{y^2}$, so the general solution is $x\sin(x^2 + y) + e^{y^2} = C$.
11	written	$\rho = e^{\frac{3}{2}x^2}(x^2 + 1)^{-3/2}$, $C = 3$: $y = e^{-\frac{3}{2}x^2}\left[-2 + 3\sqrt{(x^2 + 1)^3}\right]$.
12	written	$\det A = -4$, $A^{-1} = -\frac{1}{4}\operatorname{adj}(A)$ with rows $(1, 1, 1)$, $(4, \frac{15}{4}, \frac{13}{4})$, $(7, \frac{25}{4}, \frac{23}{4})$.